

Beam Element Section Property Tool

Requirements Document

1. Background

A long-span suspension bridge adopts a flat steel box girder as the stiffening girder. When creating the finite element model of the entire bridge, the steel box girder needs to be discretized into spatial beam elements to simulate its overall bending, torsion, shear, and warping behaviors. However, the cross-section of a steel box girder consists of top plates, bottom plates, webs, longitudinal ribs, and diaphragms, resulting in a complex geometric shape. Its effective sectional properties (such as warping constant, torsional moment of inertia, centroid offset, shear center location, etc.) cannot be directly obtained using simple formulas.

If these parameters are calculated manually, the process is not only time-consuming but also prone to neglect local stiffening effects, leading to stiffness errors that affect the accuracy of stress states and dynamic characteristic analyses of the completed bridge. Therefore, engineers need to use a beam element section property tool to accomplish the following tasks:

Section geometric modeling: Input the dimensions, thicknesses, and material properties of each plate component of the steel box girder to quickly define arbitrarily complex cross-sections.

Visual verification: Display the section shape, stress calculation points, and shear center offset direction to avoid input errors.

Output interfacing: Export the computed sectional properties directly as beam section definition cards for general-purpose finite element software (e.g., ANSYS, ABAQUS, Midas Civil).

This tool significantly improves the efficiency and reliability of detailed modeling for large bridges, ensuring the accuracy of subsequent static, stability, and wind resistance analyses.

2. Business Requirements

2.1 Overall Requirements

The section coordinate system shall adopt a global Cartesian coordinate system, following the right-hand rule.

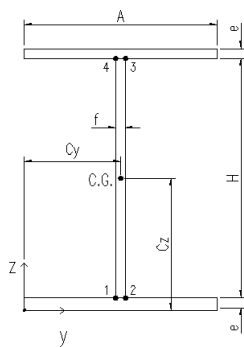
Data units and decimal places shall be standardized. The default unit is mm, with two decimal places retained.

2.2 Basic Functional Requirements

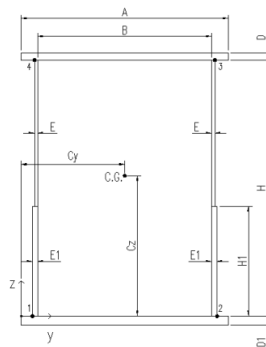
2.2.1. Section types

The section types to be developed in this phase include: **H Section**, **Box Section**, **Pipe section**, and **typical quayside crane girder sections** — 4 types in total. The meanings of the geometric dimensions and section property parameters for each section type are as follows:

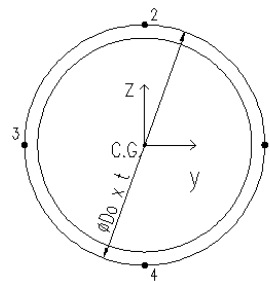
(1) Illustrations of section geometry dimensions :



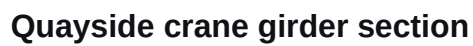
H Section



Box Section



Pipe section



AREA : Cross-sectional area

J_y : Moment of inertia about the Y axe

Jyo : Moment of inertia about the Y axe in the centroidal principal coordinate system

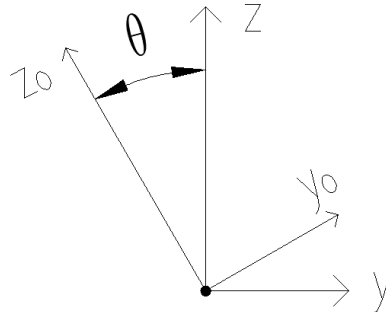
Az : Shear area in the Z direction

A_y : Shear area in the Y direction

c_z : z-coordinate of the centroid

c_y : y-coordinate of the centroid

(3) Coordinate axes



YZ : cross-sectional coordinate system

Y_0Z_0 : section centroidal principal coordinate system

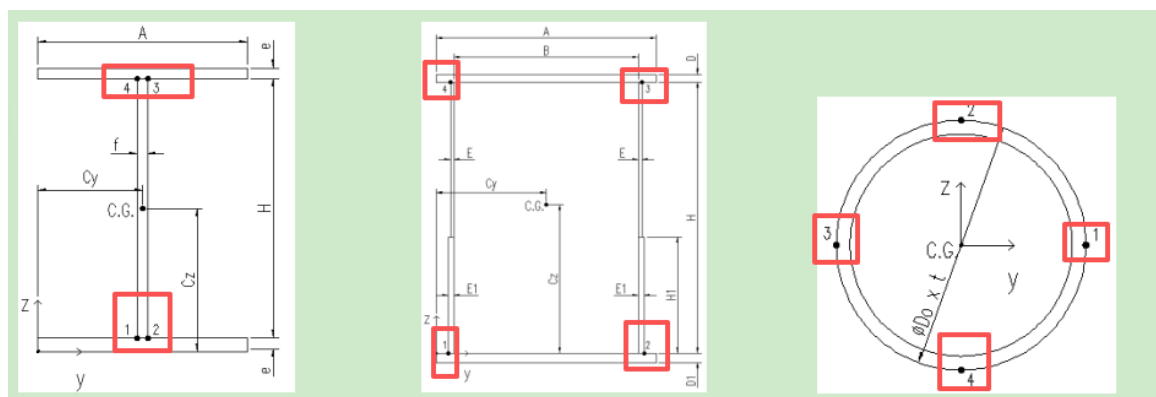
θ : Element X-axis rotation

(4) Stress points

Points 1/2/3/4 shown in the figure **are the stress output points** that need to be defined for each section in this design.

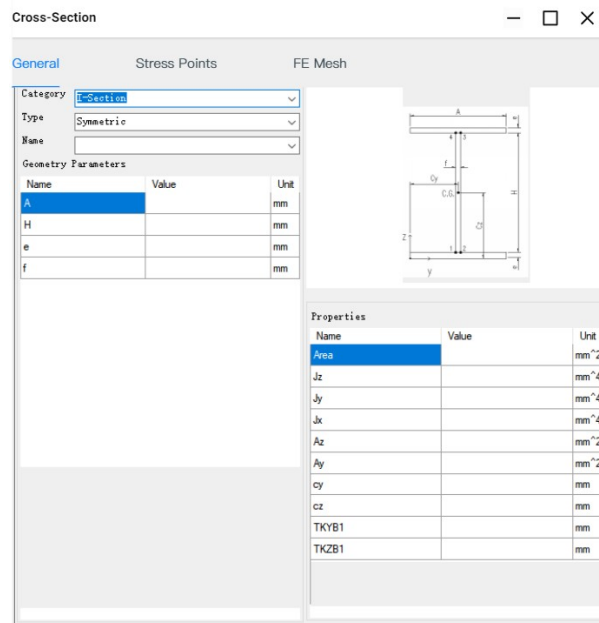
For the four specified cross-sections, the stress output points are located at the weld seams where the plates of each cross-section are joined. In engineering practice, the stress calculation results at the weld seams need to be extracted to assess weld fatigue.

The output points for the four cross-sections in this basic functional design are provided in the document, as shown in the figure below.



2.2.2. Software Interface Layout

It is mainly divided into three parts: General, Stress Points, FEMesh.



2.2.3. Main Software Functions

After the software is opened, there are three tabs: General, Stress Points, and FE Mesh. These three tabs are independent of each other, but they are internally related. General serves as the input. After the input parameters generate a cross-section, Stress Points and FE Mesh should output that cross-section synchronously, while also updating the stress output point coordinates of the cross-section and the finite element mesh diagram of the cross-section.

(1) General: Parametric input of section information. Enter the geometric dimensions of the cross-section, and the section properties will be automatically calculated and displayed. Schematic diagrams of different section types are shown below:

Cross-Section

General

Stress Points

FE Mesh

Category

H-Section

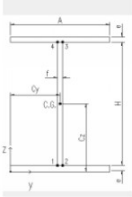
Type

Symmetric

Name

Geometry Parameters

Name	Value	Unit
A		mm
H		mm
e		mm
f		mm



Properties

Name	Value	Unit
Area		mm^2
Jz		mm^4
Jy		mm^4
Jx		mm^4
Az		mm^2
Ay		mm^2
cy		mm
cz		mm
TKYB1		mm
TKZB1		mm

H Section

Cross-Section

General

Stress Points

FE Mesh

Category

Hollow Box

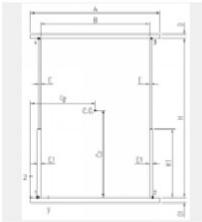
Type

Type A-Web Thickness Variation

Name

Geometry Parameters

Name	Value	Unit
A		mm
B		mm
H		mm
D		mm
E		mm
H1		mm
D1		mm
E1		mm



Properties

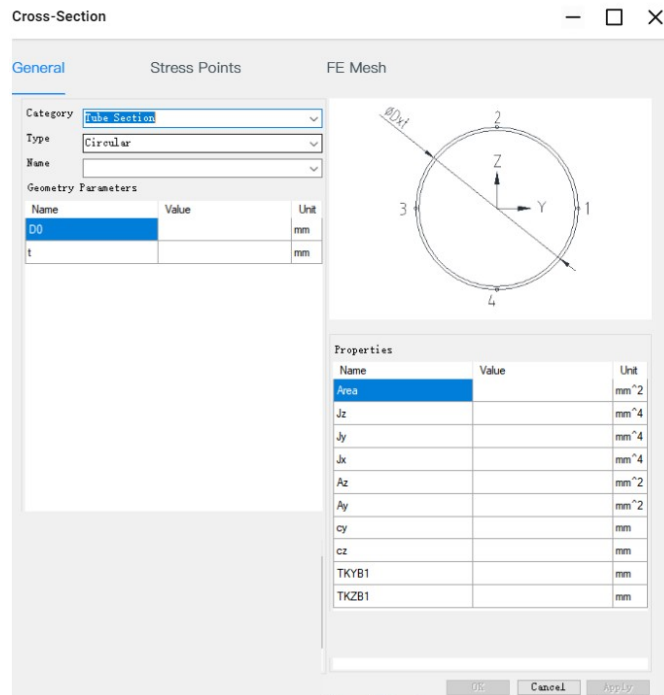
Name	Value	Unit
Area		mm^2
Jz		mm^4
Jy		mm^4
Jx		mm^4
Az		mm^2
Ay		mm^2
cy		mm
cz		mm
TKYB1		mm
TKZB1		mm

OK

Cancel

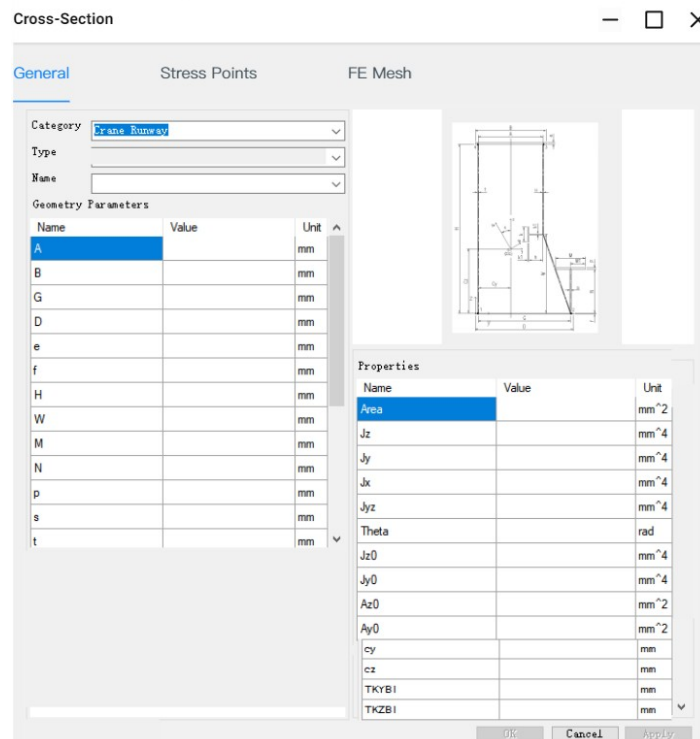
Apply

Box Section



Pipe

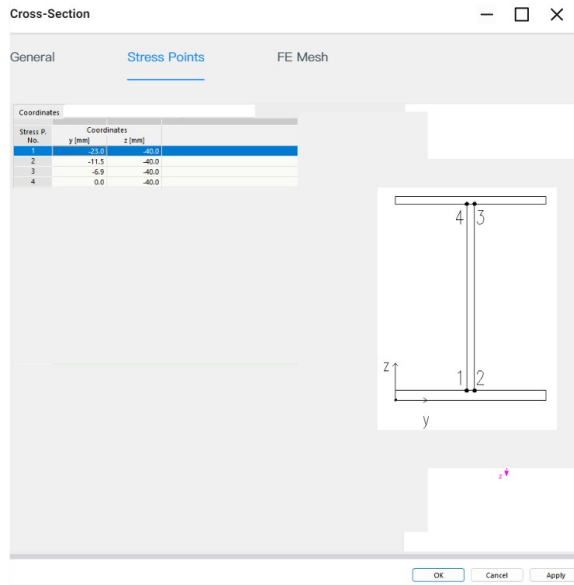
Section



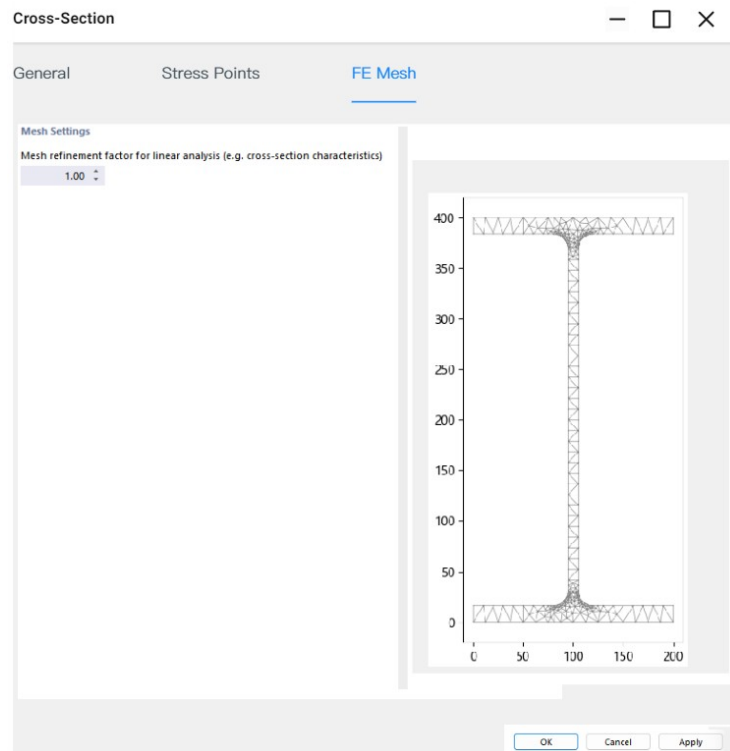
Girder Section

(2) **Stress Points:** Users can customize the coordinates of stress output

points. For the four types of sections in this design, the default stress output points are as shown in the section geometry diagram in Section 2.2. Taking the H-shaped section as an example, the software interface is shown below:



(3) FE Mesh: Mesh size setting for finite element discretization of the section. After adjusting the mesh size, the mesh division can be synchronously updated and displayed graphically. Taking the H-shaped section as an example, the software interface is shown below:



2.2.4. Advanced Functional Requirements

The program provides a canvas feature, allowing users to generate cross-sections for complex thin-walled sections through freehand drawing.

Implementation method: Users draw the centerlines of individual plate components on the canvas and create cross-sections by assigning thicknesses to these centerlines. Refer to relevant videos of Rsection for guidance.

The Canvas function is an advanced version of the General function. General provides four cross-sections (which can be extended in the future) and generates a cross-section by entering geometric parameters. In the software, it can list the stress output point coordinates (Stress Points) and the finite element mesh diagram of the cross-section (FE Mesh). The Canvas function can be understood as a feature parallel to the basic function, with the difference lying in the way the graphics are defined: one is defined by parameters, while the other is directly hand-drawn by the user.

3. Programming Development Requirements

3.1 Development Environment

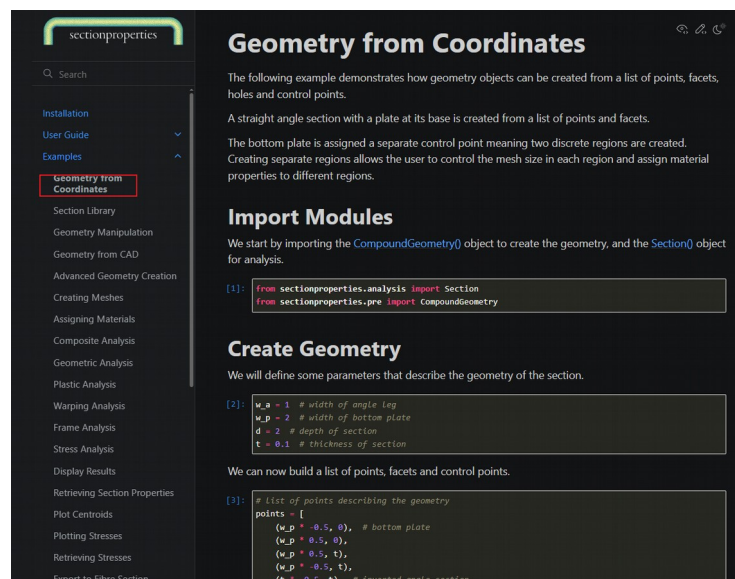
- (1) Operating system: Windows 10/11
- (2) Development language: C++17
- (3) GUI development: Qt 5.15.2

3.2 Implemented Interfaces

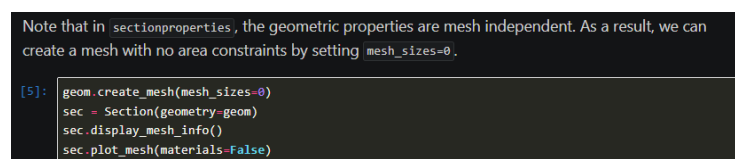
The development result will be called as a dynamic library, requiring the implementation of necessary interfaces to accommodate calls from other program modules. Interface definitions will be provided subsequently.

3.3 Recommended Python Library Interface

Geometry contour creation interface: `from_points` function.



Mesh generation interface: `create_mesh` function.



calculation

calculate_geometry_properties

```
[6]: sec.calculate_geometric_properties()
```

One easy way to view all the results that have been calculated to date, is to print them to the terminal by using `display_results()`.

```
[7]: sec.display_results(fmt=".0f")
```

Property	Value
area	3232
perimeter	911
qx	328027
qy	214915
ixx_g	56839451
Iyy_g	17355280
ixy_g	21813826
cx	66
cy	102
ixx_c	23544664
Iyy_c	3063383
ixy_c	-0
zxx+	231967
zxx-	231967
zyy+	46066
zyy-	46066
rx	85
ry	31
i11_c	23544664
i22_c	3063383
phi	0
z11+	231967
z11-	231967
z22+	46066
z22-	46066
r11	85
r22	31

All interfaces and usage examples can be found in the help documentation.

https://sectionproperties.readthedocs.io/en/stable/examples/analysis/geometric_analysis.html#

3.4 Code Requirements

Main deliverables to be submitted:

- (1) A self-compilable DLL (dynamic-link library).
- (2) Examples demonstrating the use of the DLL.
- (3) Clear interface documentation.

Folder directory structure:

Section_Property_Tool

```
-- Source/

    -- dir|*.cpp|*.cxx|*.h|*.hpp

-- Library/

    -- bin/

    -- lib/

    -- include/

-- Examples/

    -- example1/

        -- main.cpp

        -- ...

    -- example2/

        -- main.cpp

        -- ...

-- Documents/

    -- User Manual.docx

    -- README.md
```

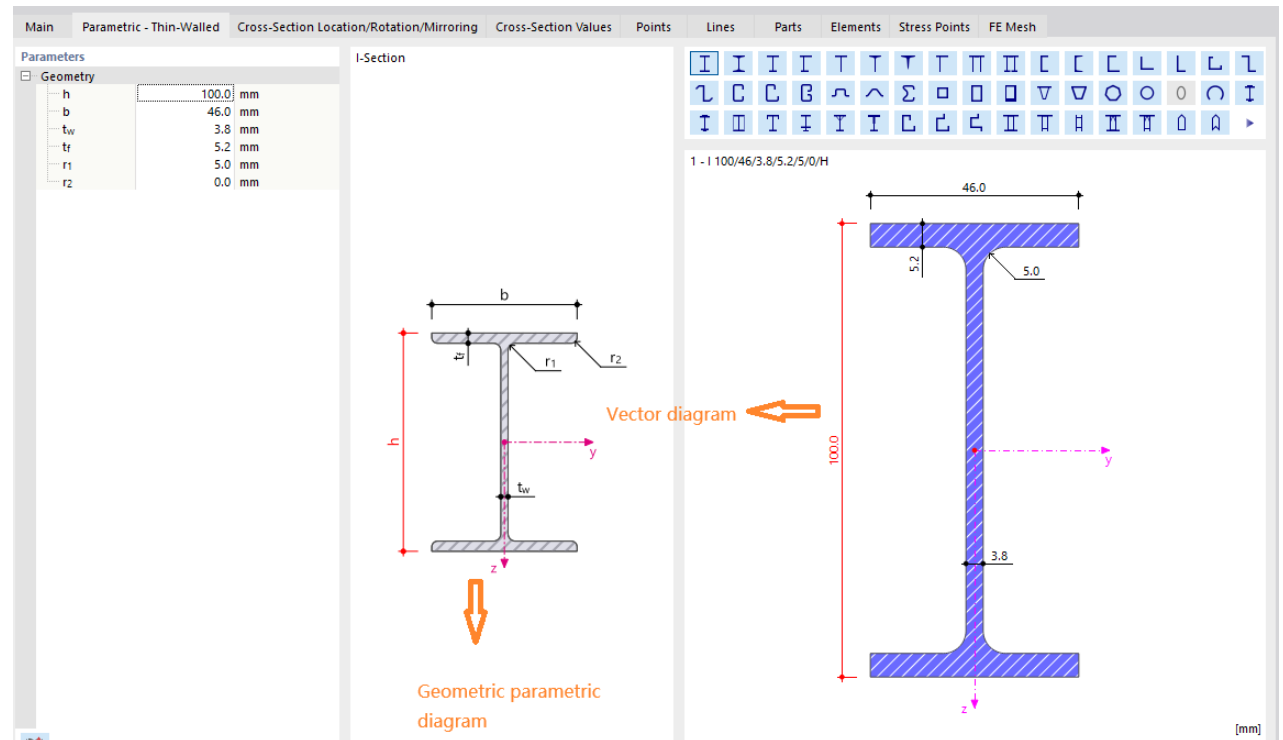
4. Commercial Software Reference

RSECTION is a professional section property and stress analysis software developed by Dlubal Software GmbH, Germany, for reference. The software provides a 90-day trial version, and the download link is as follows:

<https://www.dlubal.com/en/downloads>

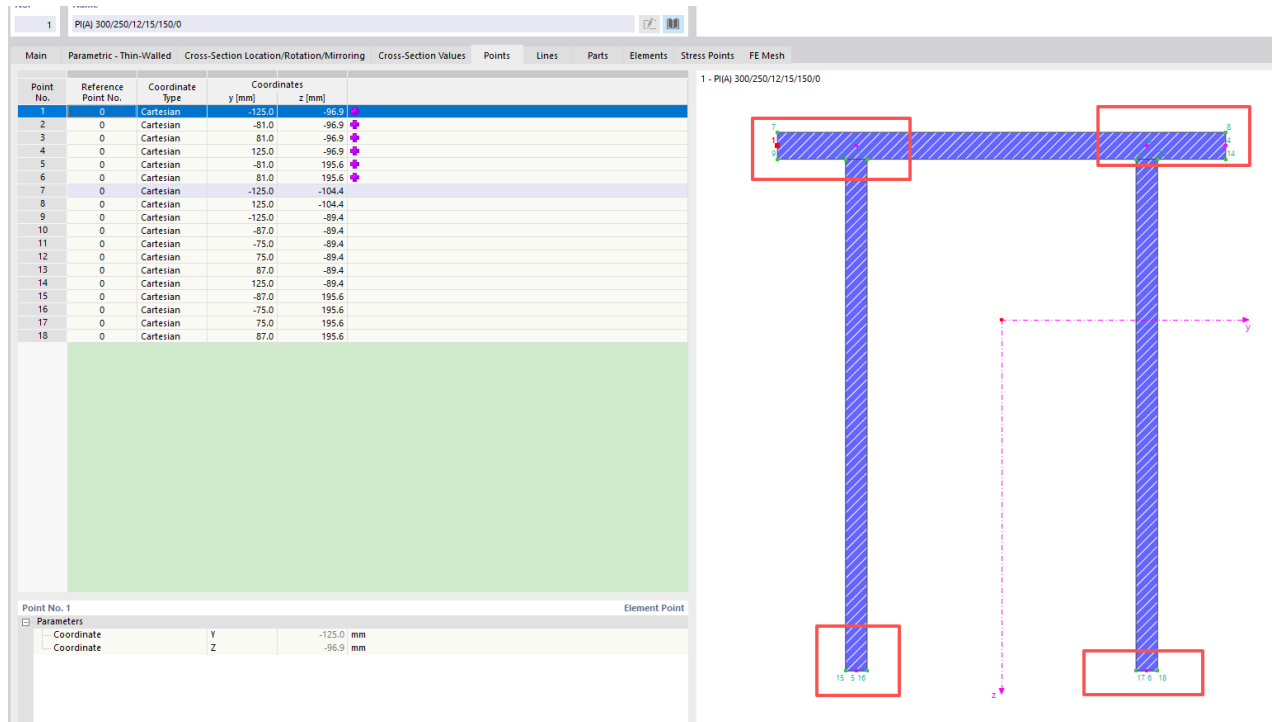
Example1 screen from RSECTION

In the Rsection software, this window provides two geometric figures: one is an image illustrating the geometric dimensions, and the other is a vector graphic that changes with the parameters, as shown in the figure below.



Example2 screen from RSECTION

For a custom cross-section, i.e., the points where the various plates intersect, the Rsection software provides a relatively large number of stress output points, as shown in the figure below.



5. Submission Materials

5.1. Requirements and Design Documents

Architecture design document: module division (geometry engine, calculation core, GUI, input/output interfaces, AI integration points), class diagrams, data flow diagrams.

5.3. Development Artifacts

Complete source code: hosted in a Git repository (provide link or archive) with a clear directory structure and a README.md (build/run instructions).

Configuration files: dependency list (e.g., requirements.txt for C++), build scripts.

5.3. Testing and Quality Assurance

Test report: includes test cases (input dimensions, expected output, actual output), performance tests (time consumption for processing complex cross-sections), accuracy comparison (against commercial software or

analytical solutions).

Defect tracking log: bugs found, fix records, AI-assisted debugging process.

5.4. User Documentation

User manual: installation steps, GUI function descriptions, section input methods (parametric or graphical drawing), result interpretation.

Example tutorials: complete operation demonstrations for 2–3 typical engineering sections (e.g., rectangular, I-shaped, box-shaped).

5.5. Project Summary and Presentation

Project summary report: workload statistics, benefits and limitations of AI assistance, reflections on team collaboration, similarities and differences between industrial software development and general software engineering.

Final presentation slides: background, function demonstration, key technical points, AI-assisted cases, test results.

Demonstration video (1–3 minutes): core operational workflow for acceptance.